BS-M301 — Probability and Statistics

Semester III • 3L:0T:3P • 3 Credits

Module 4 Exam Notes: Basic Statistics, Moments, Distributions, Correlation and Regression

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1 Measures of Central Tendency, Moments, Skewness and Kurtosis 2
1.1 Moments and Moment Conversion Relations 2
1.2 Measures of Skewness (Asymmetry) 3
1.3 Measures of Kurtosis (Peakedness) 3
2 Evaluation of Statistical Parameters for Core Distributions 4
2.1 Binomial Distribution: Parameters and Recurrence Relation 4
2.2 Poisson Distribution: Parameters and Recurrence Relation 4
2.3 Normal Distribution: Parameters and Recurrence Relation 4
3 Correlation and Regression Analysis 5
3.1 Karl Pearson's Coefficient of Correlation (r) 5
3.2 Linear Regression Equations and Least Squares 5
3.3 Angle Between the Two Regression Lines 6
3.4 Spearman's Rank Correlation Coefficient (r_s) 7

1. Measures of Central Tendency, Moments, Skewness and Kurtosis

Descriptive statistics provides quantitative summaries describing the central clustering, dispersion, asymmetry, and peakedness of experimental datasets.

Definition: Measures of Central Tendency

1. **Arithmetic Mean ($\bar{x}$):** The sum of all observations divided by sample size:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{n} f_i x_i, \qquad N = \sum_{i=1}^{n} f_i \tag{1}$$

Key Properties: (1) Algebraic sum of deviations is zero: $\sum f_i(x_i - \bar{x}) = 0$. (2) Sum of squared deviations is minimal about the mean: $\sum f_i(x_i - A)^2$ is minimum when $A = \bar{x}$.

2. **Median (M):** The middle value partitioning the ordered dataset into two equal halves. For grouped data with median class $[L, L+h]$:

$$M = L + \frac{\frac{N}{2} - C}{f} \cdot h \tag{2}$$

where L is lower boundary, C is cumulative frequency of preceding class, and f is frequency.

3. **Mode (Z):** The most frequently occurring value in the distribution:

$$Z = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \cdot h \tag{3}$$

Empirical Relationship for Moderately Skewed Distributions:

$$\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean} \iff \text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median}) \tag{4}$$

1.1 Moments and Moment Conversion Relations

- **r-th Raw Moment about Origin (μ'_r):** $\mu'_r = \frac{1}{N}\sum f_i x_i^r$. Note: $\mu'_1 = \bar{x}$.
- **r-th Central Moment about Mean (μ_r):** $\mu_r = \frac{1}{N}\sum f_i(x_i - \bar{x})^r$.

Conversion Relations: Central Moments from Raw Moments

Expanding $(x_i - \bar{x})^r = \sum_{k=0}^{r} \binom{r}{k} x_i^{r-k}(-\bar{x})^k$:

$$\mu_1 = 0 \quad \text{(Always zero!)} \tag{5}$$
$$\mu_2 = \mu'_2 - (\mu'_1)^2 = \sigma^2 \quad \text{(Variance)} \tag{6}$$
$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 \tag{7}$$
$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 \tag{8}$$

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1.2 Measures of Skewness (Asymmetry)

Skewness quantifies the departure of a distribution curve from perfect bilateral symmetry.

Formulas for Coefficients of Skewness

1. **Karl Pearson's Coefficient of Skewness (S_k):**

$$S_k = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{3(\text{Mean} - \text{Median})}{\sigma}, \qquad -3 \le S_k \le 3 \tag{9}$$

2. **Bowley's Quartile Coefficient of Skewness (S_b):** Based on quartiles $Q_1, Q_2(\text{Median}), Q_3$:

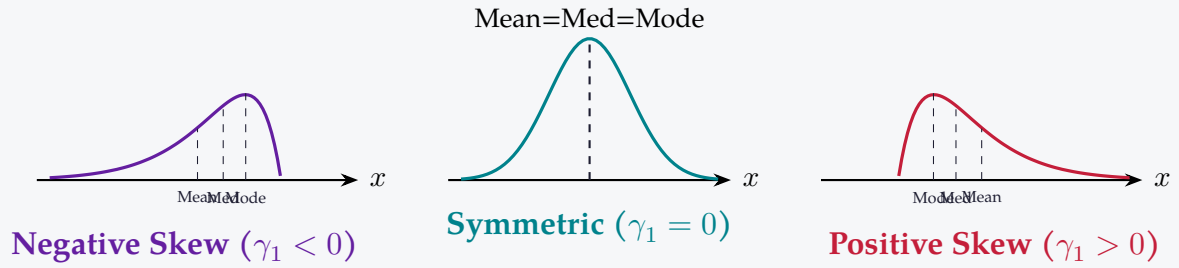
$$S_b = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}, \qquad -1 \le S_b \le 1 \tag{10}$$

3. **Moment Coefficient of Skewness (β_1 and γ_1):**

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\mu_3^2}{\sigma^6}, \qquad \gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\sigma^3} \tag{11}$$

- $\gamma_1 = 0$: Symmetric distribution (Mean = Median = Mode).
- $\gamma_1 > 0$: Positively skewed (long tail stretched to the right; Mean > Median > Mode).
- $\gamma_1 < 0$: Negatively skewed (long tail stretched to the left; Mode > Median > Mean).

TikZ Schematic: Geometric Comparison of Symmetrical and Skewed Distributions



Negative Skew ($\gamma_1 < 0$) Symmetric ($\gamma_1 = 0$) Positive Skew ($\gamma_1 > 0$)

1.3 Measures of Kurtosis (Peakedness)

Definition: Kurtosis Coefficients β_2 and γ_2

Kurtosis quantifies the degree of peakedness and the heaviness of the tails of a frequency curve relative to the standard normal bell curve:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}, \qquad \gamma_2 = \beta_2 - 3 \tag{12}$$

- **Mesokurtic ($\beta_2 = 3, \gamma_2 = 0$):** Normal distribution curve profile.
- **Leptokurtic ($\beta_2 > 3, \gamma_2 > 0$):** Sharply peaked center with heavy, fat tails (e.g., Student's t, Laplace).

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- **Platykurtic** ($\beta_2 < 3, \gamma_2 < 0$)**:** Flatter plateau-like peak with light, rapidly vanishing tails (e.g., Uniform distribution where $\beta_2 = 1.8$).

2. Evaluation of Statistical Parameters for Core Distributions

2.1 Binomial Distribution: Parameters and Recurrence Relation

For $X \sim \text{Bin}(n, p)$ with $q = 1 - p$:

$$M_X(t) = \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k q^{n-k} = (q + pe^t)^n \tag{13}$$

$$E[X] = \mu = np, \qquad \text{Var}(X) = \sigma^2 = npq \tag{14}$$

$$\mu_3 = npq(q - p), \qquad \mu_4 = 3n^2p^2q^2 + npq(1 - 6pq) \tag{15}$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(q-p)^2}{npq}, \qquad \beta_2 = \frac{\mu_4}{\mu_2^2} = 3 + \frac{1 - 6pq}{npq} \tag{16}$$

Recurrence Relation for Central Moments:

$$\mu_{r+1} = pq \left[nr\mu_{r-1} + \frac{d\mu_r}{dp} \right] \tag{17}$$

2.2 Poisson Distribution: Parameters and Recurrence Relation

For $X \sim \text{Poisson}(\lambda)$:

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = \exp\left(\lambda(e^t - 1)\right) \tag{18}$$

$$E[X] = \lambda, \qquad \text{Var}(X) = \lambda \tag{19}$$

$$\mu_3 = \lambda, \qquad \mu_4 = 3\lambda^2 + \lambda \tag{20}$$

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\lambda^2}{\lambda^3} = \frac{1}{\lambda}, \qquad \beta_2 = \frac{3\lambda^2 + \lambda}{\lambda^2} = 3 + \frac{1}{\lambda} \tag{21}$$

Recurrence Relation for Central Moments:

$$\mu_{r+1} = \lambda \left[r\mu_{r-1} + \frac{d\mu_r}{d\lambda} \right] \tag{22}$$

As $\lambda \to \infty$, both $\beta_1 \to 0$ and $\beta_2 \to 3$, confirming convergence of Poisson to Normal!

2.3 Normal Distribution: Parameters and Recurrence Relation

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$E[X] = \mu, \qquad \text{Var}(X) = \sigma^2 \tag{23}$$

$$\mu_{2k+1} = 0 \quad \forall k \in \mathbb{N}_0, \qquad \mu_{2k} = \frac{(2k)!}{2^k k!} \sigma^{2k} \tag{24}$$

$$\beta_1 = 0, \qquad \beta_2 = \frac{3\sigma^4}{\sigma^4} = 3 \iff \gamma_2 = 0 \tag{25}$$

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Recurrence Relation for Even Central Moments:

$$\mu_{2k} = (2k-1)\sigma^2\mu_{2k-2} \tag{26}$$

3. Correlation and Regression Analysis

3.1 Karl Pearson's Coefficient of Correlation (r)

Karl Pearson's product-moment correlation coefficient r provides a scale-invariant measure of linear association between two paired variables (x_i, y_i):

Formula for Pearson's Correlation Coefficient

$$r = \frac{\mathrm{Cov}(X,Y)}{\sigma_x\sigma_y} = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}} = \frac{n\sum x_iy_i - (\sum x_i)(\sum y_i)}{\sqrt{n\sum x_i^2-(\sum x_i)^2}\sqrt{n\sum y_i^2-(\sum y_i)^2}} \tag{27}$$

Properties of r:

1. Normalized limits: $-1 \le r \le +1$.
2. Invariant to change of origin and scale: If $u = \frac{x-a}{h}$ and $v = \frac{y-b}{k}$ with $h, k > 0$, then $r_{xy} = r_{uv}$.

3.2 Linear Regression Equations and Least Squares

Regression expresses the average functional relationship predicting a dependent variable from an independent variable.

The Two Lines of Regression

1. Line of Regression of Y on X: Predicts Y for a known value of X:

$$Y - \bar{y} = b_{yx}(X-\bar{x}), \qquad b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{\mathrm{Cov}(X,Y)}{\sigma_x^2} \tag{28}$$

2. Line of Regression of X on Y: Predicts X for a known value of Y:

$$X - \bar{x} = b_{xy}(Y-\bar{y}), \qquad b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_y^2} \tag{29}$$

where b_{yx} and b_{xy} are the **regression coefficients**.

Key Properties of Regression Coefficients

1. **Geometric Mean Property:** The correlation coefficient is the geometric mean of the

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regression coefficients:

$$r = \pm\sqrt{b_{yx} \cdot b_{xy}} \tag{30}$$

Both regression coefficients must share the same algebraic sign, which is also the sign of r.

2. **Reciprocal Magnitude Constraint:** If $b_{yx} > 1$, then $b_{xy} < 1$, since $b_{yx}b_{xy} = r^2 \leq 1$.
3. **Arithmetic Mean Inequality:** $\dfrac{b_{yx} + b_{xy}}{2} \geq r$ (by AM $\geq$ GM).
4. **Centroid Intersection:** Both lines intersect at the point of means $(\bar{x}, \bar{y})$.
5. **Scale Dependence:** Regression coefficients are invariant to change of origin, but **depend** on scale: $b_{yx} = \frac{k}{h} b_{vu}$.

3.3 Angle Between the Two Regression Lines

Derivation: Angle θ Between Regression Lines

The slopes of the two lines in the (X, Y) plane are:

$$m_1 = b_{yx} = r\frac{\sigma_y}{\sigma_x}, \qquad m_2 = \frac{1}{b_{xy}} = \frac{\sigma_y}{r\sigma_x} \tag{31}$$

The angle θ between the lines satisfies:

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{\frac{\sigma_y}{r\sigma_x} - r\frac{\sigma_y}{\sigma_x}}{1 + \left(r\frac{\sigma_y}{\sigma_x}\right)\left(\frac{\sigma_y}{r\sigma_x}\right)}\right| = \left|\frac{\frac{\sigma_y}{\sigma_x}\left(\frac{1-r^2}{r}\right)}{1 + \frac{\sigma_y^2}{\sigma_x^2}}\right| \tag{32}$$

Simplifying the denominator $1 + \sigma_y^2/\sigma_x^2 = (\sigma_x^2 + \sigma_y^2)/\sigma_x^2$:

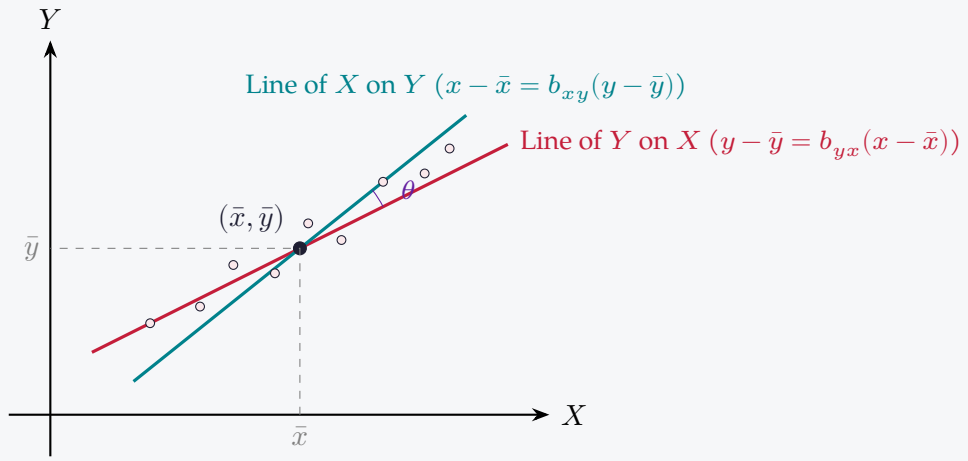
$$\tan\theta = \left(\frac{1 - r^2}{|r|}\right)\left(\frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}\right) \tag{33}$$

Special Cases:

- If $r = \pm 1 \implies \tan\theta = 0 \implies \theta = 0^\circ$ (the two lines **coincide**; perfect linear correlation).
- If $r = 0 \implies \tan\theta \to \infty \implies \theta = 90^\circ$ (the two lines are **perpendicular** parallel to coordinate axes; uncorrelated).

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TikZ Plot: Scatter Diagram and the Two Lines of Regression with Intersection at Centroid



3.4 Spearman's Rank Correlation Coefficient (r_s)

When variables represent qualitative attributes that cannot be measured numerically but can be ordered or ranked (e.g., beauty, intelligence, performance rankings):

Spearman's Rank Correlation Formula

Let R_{xi} and R_{yi} denote the ranks assigned to the i-th individual, with difference $d_i = R_{xi} - R_{yi}$.

1. **Case 1: When Ranks are Distinct (No Ties):**

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)} \tag{34}$$

2. **Case 2: When Ranks are Tied (Correction Factor):** If an attribute value repeats m times, all tied entries are allocated the average rank, and a correction factor $\frac{m(m^2-1)}{12}$ is added for each tie group:

$$r_s = 1 - \frac{6\left[\sum_{i=1}^{n} d_i^2 + \sum_{j} \frac{m_j(m_j^2-1)}{12}\right]}{n(n^2-1)} \tag{35}$$

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Quick Revision — Module 4

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Key Formulas and Statistical Identities at a Glance

Empirical Mean-Mode	$\text{Mode} \approx 3\text{Median} - 2\text{Mean} \iff \text{Mean} - \text{Mode} \approx 3(\text{Mean} - \text{Median})$
Central from Raw Moments	$\mu_2 = \mu_2' - (\mu_1')^2,\ \mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3,\ \mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$
Pearson Skewness	$S_k = \frac{\text{Mean}-\text{Mode}}{\sigma} = \frac{3(\text{Mean}-\text{Median})}{\sigma},\quad \beta_1 = \frac{\mu_3^2}{\mu_2^3},\quad \gamma_1 = \frac{\mu_3}{\sigma^3}$
Kurtosis Coefficients	$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4},\quad \gamma_2 = \beta_2 - 3\quad (\text{Meso} = 3, \text{Lepto} > 3, \text{Platy} < 3)$
Binomial Parameters	$\mu = np,\quad \sigma^2 = npq,\quad \beta_1 = \frac{(q-p)^2}{npq},\quad \beta_2 = 3 + \frac{1-6pq}{npq}$
Poisson Parameters	$\mu = \lambda,\quad \sigma^2 = \lambda,\quad \beta_1 = \frac{1}{\lambda},\quad \beta_2 = 3 + \frac{1}{\lambda}$
Normal Moments	$\text{All odd } \mu_{2k+1} = 0,\quad \mu_{2k} = \frac{(2k)!}{2^k k!}\sigma^{2k},\quad \beta_1 = 0,\quad \beta_2 = 3$
Pearson Correlation	$r = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}} \in [-1, +1]$
Regression Lines	$Y - \bar{y} = b_{yx}(X - \bar{x}),\quad X - \bar{x} = b_{xy}(Y - \bar{y}),\quad b_{yx} = r\frac{\sigma_y}{\sigma_x},\ b_{xy} = r\frac{\sigma_x}{\sigma_y}$
Regression Properties	$r = \pm\sqrt{b_{yx} b_{xy}},\quad \frac{b_{yx}+b_{xy}}{2} \geq r,\quad \tan\theta = \frac{1-r^2}{\lvert r\rvert}\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$
Spearman Rank Corr.	$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}\quad (\text{Ties add } \sum \frac{m_j(m_j^2-1)}{12})$

One-Line Definitions

- **Arithmetic Mean:** Center of gravity of data where algebraic sum of deviations $\sum(x_i - \bar{x})$ is identically zero.
- **Median & Mode:** Median is positional 50th percentile; Mode is observation exhibiting peak frequency density.
- **Skewness:** Metric quantifying directional asymmetry of frequency curve tail relative to central peak.
- **Kurtosis:** Measure of peakedness and tail fatness relative to standard Gaussian benchmark ($\beta_2 = 3$).
- **Leptokurtic Curve:** Sharply peaked frequency profile with fat heavy tails exhibiting $\beta_2 > 3$.
- **Platykurtic Curve:** Broad-shouldered flattened frequency curve with light rapidly decaying tails ($\beta_2 < 3$).
- **Pearson's r:** Standardized dimensionless measure of the strength and direction of linear association.
- **Regression Line:** Optimal line minimizing sum of squared vertical/horizontal prediction errors (Least Squares).
- **Regression Coefficients:** Directional slopes b_{yx} and b_{xy} whose geometric mean equals correlation coefficient r.
- **Spearman's Rank Correlation:** Non-parametric correlation measuring monotonic agreement between ordinal rankings.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the algebraic properties of the Arithmetic Mean.

(1) The sum of deviations of all observations from their arithmetic mean is zero: $\sum f_i(x_i - \bar{x}) = 0$. (2) The sum of squared deviations is minimal when taken about the mean: $\sum f_i(x_i - A)^2$ is minimum for $A = \bar{x}$. (3) Mean of combined series: $\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1+n_2}$.

Q2. Derive the relationship $\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$.

$\mu_3 = E[(X-\mu)^3] = E[X^3 - 3X^2\mu + 3X\mu^2 - \mu^3] = E[X^3] - 3\mu E[X^2] + 3\mu^2 E[X] - \mu^3$. Substituting $E[X] = \mu'_1, E[X^2] = \mu'_2, E[X^3] = \mu'_3$: $\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 3(\mu'_1)^3 - (\mu'_1)^3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$.

Q3. Distinguish between positive skewness and negative skewness with sketches.

In positive skewness, the tail is extended to the right, and Mean $>$ Median $>$ Mode ($\gamma_1 > 0$). In negative skewness, the tail is extended to the left, and Mode $>$ Median $>$ Mean ($\gamma_1 < 0$).

Q4. What are Leptokurtic, Mesokurtic and Platykurtic curves?

A curve with $\beta_2 = 3$ is Mesokurtic (Normal). If $\beta_2 > 3$, it is Leptokurtic (more peaked with heavier tails). If $\beta_2 < 3$, it is Platykurtic (flatter top with lighter tails).

Q5. State the recurrence relation for central moments of a Poisson distribution.

$\mu_{r+1} = \lambda\left[r\mu_{r-1} + \frac{d\mu_r}{d\lambda}\right]$. For $r = 1$: $\mu_2 = \lambda(1 \cdot 0 + 1) = \lambda$. For $r = 2$: $\mu_3 = \lambda(2 \cdot 0 + \frac{d\lambda}{d\lambda}) = \lambda$. For $r = 3$: $\mu_4 = \lambda(3\lambda + \frac{d\lambda}{d\lambda}) = 3\lambda^2 + \lambda$.

Q6. Prove that the correlation coefficient is the geometric mean of the two regression coefficients.

$b_{yx} = r\frac{\sigma_y}{\sigma_x}$ and $b_{xy} = r\frac{\sigma_x}{\sigma_y}$. Multiplying: $b_{yx} \cdot b_{xy} = \left(r\frac{\sigma_y}{\sigma_x}\right)\left(r\frac{\sigma_x}{\sigma_y}\right) = r^2$. Taking the square root: $r = \pm\sqrt{b_{yx} \cdot b_{xy}}$.

Q7. Can two regression coefficients be 1.5 and 0.8? Explain.

Yes! $b_{yx} \cdot b_{xy} = (1.5)(0.8) = 1.2 > 1$. But wait: $r^2 = b_{yx}b_{xy} \leq 1$. Since $1.2 > 1$, this would imply $r^2 > 1$, which violates the Cauchy-Schwarz bound $|r| \leq 1$. Hence $b_{yx} = 1.5$ and $b_{xy} = 0.8$ is impossible!

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Frequently Asked / Viva Questions (Part 2)

Q8. Why are there two lines of regression rather than a single line?

Because the regression of Y on X minimizes the sum of squared vertical errors $\sum(y_i - \hat{y}_i)^2$, whereas the regression of X on Y minimizes the sum of squared horizontal errors $\sum(x_i - \hat{x}_i)^2$. The two lines coincide if and only if $r = \pm 1$ (zero error).

Q9. State the formula for the angle θ between the two regression lines and interpret the extreme cases.

$\tan\theta = \frac{1-r^2}{|r|}\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$. If $r = \pm 1$, $\tan\theta = 0 \implies \theta = 0^\circ$ (lines coincide; perfect correlation). If $r = 0$, $\tan\theta = \infty \implies \theta = 90^\circ$ (lines are perpendicular; uncorrelated).

Q10. Derive Spearman's Rank Correlation formula when ranks are not tied.

Let x_i, y_i be ranks from 1 to n. Then $\bar{x} = \bar{y} = \frac{n+1}{2}$, $\sigma_x^2 = \sigma_y^2 = \frac{n^2-1}{12}$. Since $d_i = x_i - y_i$, $\sum d_i^2 = \sum(x_i-\bar{x})^2 + \sum(y_i-\bar{y})^2 - 2\sum(x_i-\bar{x})(y_i-\bar{y}) = 2n\sigma_x^2 - 2nr\sigma_x^2 = 2n\sigma_x^2(1-r_s)$. Substituting $\sigma_x^2 = \frac{n^2-1}{12}$ yields $r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$.

Summary Matrix: Comparison of Binomial, Poisson and Normal Distributions

Parameter	**Binomial** $\mathbf{Bin}(n,p)$	**Poisson** $\mathbf{Poisson}(\lambda)$	**Normal** $\mathcal{N}(\mu,\sigma^2)$	**Comparative Property**
Mean ($E[X]$)	np	λ	μ	Location measure.
Variance (σ^2)	$npq < \mu$	$\lambda = \mu$	σ^2 **(Independent of** μ**)**	$\text{Var}_{\text{Bin}} < \text{Mean}$; $\text{Var}_{\text{Poi}} = \text{Mean}$.
MGF ($M_X(t)$)	$(q+pe^t)^n$	$\exp(\lambda(e^t - 1))$	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$	Poisson is limiting Binomial MGF.
Skewness (γ_1)	$\frac{q-p}{\sqrt{npq}}$	$\frac{1}{\sqrt{\lambda}}$	0	Binomial is symmetric if $p = 0.5$.
Kurtosis (β_2)	$3 + \frac{1-6pq}{npq}$	$3 + \frac{1}{\lambda}$	3 ($\gamma_2 = 0$)	Both converge to 3 as $n, \lambda \to \infty$.

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